MYSQL MODULE CAPSTONE PROJECT

E-Learning Database Management & Data Analysis

1. Executive Summary

This capstone project demonstrates the design and use of a relational database for an E-Learning platform using MySQL. The database contains three connected tables—learners, courses, and purchases—and uses primary-key and foreign-key constraints to represent relationships between learner records, course records, and purchase transactions. SQL queries are used to examine learner spending, course popularity, category revenue, purchasing behavior, and courses with no recorded purchases.

2. Project Overview

The project creates a database named elearning and inserts sample records into its three tables. The analysis uses joins, grouping, aggregate functions, filtering, and sorting to answer practical questions about course purchases.

Project Objectives

- Create an E-Learning database with related learner, course, and purchase tables.
- Define primary keys and foreign keys to establish table relationships.
- Retrieve information by joining data across multiple tables.
- Calculate learner-level spending and course/category-level purchase metrics.
- Identify learners purchasing across multiple course categories.
- Find courses that do not appear in the purchase records.

3. Tools and Technologies

Technology	Purpose in the project
MySQL	Relational database creation and SQL-based analysis
SQL	Data retrieval, joins, aggregation, filtering, and sorting
MySQL Workbench	Database environment for writing and executing SQL (as the intended workbench tool)

4. Database Design

The SQL script creates the database elearning and defines the following three tables:

learners — Stores learner_id (primary key), full_name, and country.

courses — Stores course_id (primary key), course_name, category, and unit_price.

purchases — Stores purchase_id (primary key), learner_id, course_id, quantity, and purchase_date. learner_id and course_id are foreign keys referencing learners and courses.

Relationship Overview

The purchases table connects learners and courses. Each purchase record references a learner and a course through foreign keys. This structure allows purchase information to be analyzed alongside learner and course details.

5. Sample Data Included

The supplied SQL script inserts 5 learner records, 5 course records, and 8 purchase records. The sample learners are from India, the USA, and the UK. The courses cover Programming, Database, and Analytics categories.

Table	Number of sample records	Main information
learners	5	Learner name and country
courses	5	Course, category, and unit price
purchases	8	Learner/course references, quantity, and date

6. SQL Concepts Applied

- CREATE DATABASE and CREATE TABLE — database and table creation.
- PRIMARY KEY and FOREIGN KEY — defining unique identifiers and table relationships.
- INSERT INTO — adding sample records.
- INNER JOIN — retrieving matching purchase, learner, and course details.
- LEFT JOIN — retaining all learners while showing any matching purchases.
- RIGHT JOIN — retaining all courses while showing matching purchase information.
- GROUP BY and aggregate functions — summarizing spending, quantities, revenue, and learner counts.
- HAVING — filtering grouped learner results by the number of categories purchased.
- ORDER BY and LIMIT — ranking results and selecting the top three courses.
- COUNT(DISTINCT ...) — counting unique learners or categories.

7. Analysis Questions and Query Approach

Learner purchase details

Uses INNER JOIN across purchases, learners, and courses to display learner name, course, category, quantity, calculated line amount, and purchase date. Results are ordered by total amount in descending order.

Learners' total spending

Uses LEFT JOIN and SUM(quantity × unit_price), grouped by learner. This approach includes learners even if they have no matching purchase record.

Top 3 purchased courses

Groups purchase quantities by course, sorts the total quantity in descending order, and returns the first three rows using LIMIT 3.

Category revenue and unique learners

Groups transactions by course category, calculates revenue using quantity × unit_price, and counts distinct learner IDs.

Learners purchasing from multiple categories

Counts distinct course categories for each learner and uses HAVING to retain learners with more than one category.

Courses not purchased

Uses a LEFT JOIN from courses to purchases and filters for NULL purchase IDs to find courses without a matching purchase record.

8. Key Findings Supported by the SQL Script

The SQL file defines the analysis needed to identify spending by learner, the three highest courses by purchased quantity, revenue by category, learners who purchase across categories, and courses without purchases. The script itself contains the queries but does not include captured query-output tables or a separate results section. Therefore, this report describes the analysis performed by the queries without asserting numerical findings that are not shown in the supplied file.

9. Conclusion

The MySQL Module Capstone Project provides hands-on practice in relational database design and SQL-based analysis. By creating linked tables and applying joins, aggregate functions, grouping, filtering, and sorting, the project demonstrates how transaction data can be organized and queried to answer useful E-Learning business questions. It strengthens practical skills in MySQL, database relationships, and analytical query development.

10. Skills Demonstrated

MySQL • SQL Query Writing • Relational Database Design • Primary and Foreign Keys • INNER JOIN • LEFT JOIN • RIGHT JOIN • Aggregate Functions • GROUP BY • HAVING • ORDER BY • LIMIT • Data Analysis

Project Source

This report is based on the uploaded file, “MySQL Module Capstone Project.sql”. It documents the schema, sample data, and SQL statements present in that file.